

ACADIANA, LA, USA

WMO: 722314

Lat: 30.038N

Lon: 91.884W

Elev: 7

StdP: 101.24

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	1.1	-7.9	1.9	3.5	-5.3	2.4	5.6	11.1	14.5	10.5	13.4	3.3	0	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	34.8	25.5	33.8	25.4	32.9	25.2	27.3	31.3	26.9	30.9	26.5	30.6	2.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	21.8	28.6	26.0	21.4	28.5	25.3	20.5	28.4	85.8	31.2	83.8	30.9	82.1	30.5	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.5	7.7	-3.9	36.6	1.9	1.6	-5.3	37.8	-6.4	38.7	-7.5	39.6	-8.9	40.7		
			-5.1	27.9	2.0	0.6	-6.5	28.3	-7.6	28.6	-8.7	29.0	-10.2	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	11.6	13.9	17.2	20.7	24.6	27.4	28.1	28.3	26.4	21.5	16.2	12.7
	DBStd	7.09	5.52	5.22	4.37	3.45	2.68	1.61	1.33	1.47	2.25	3.96	4.68	5.25
	HDD10.0	109	47	21	4	0	0	0	0	0	0	0	7	31
	HDD18.3	747	217	141	73	18	1	0	0	0	0	17	94	187
	CDD10.0	4037	97	128	228	323	452	522	562	492	358	193	116	
	CDD18.3	1633	8	15	39	91	195	272	303	308	242	116	30	13
	CDH23.3	15112	18	44	187	638	1728	2748	3172	3243	2157	951	179	46
	CDH26.7	5852	0	1	13	107	597	1184	1352	1431	864	286	18	1
Wind	WSAvg	3.3	4.0	4.2	4.0	4.1	3.5	2.7	2.1	2.2	2.6	2.8	3.3	3.8
Precipitation	PrecAvg	1562	143	111	100	126	127	172	153	146	134	119	108	122
	PrecMax	2167	339	232	270	303	342	455	236	708	262	352	350	342
	PrecMin	1027	20	18	8	12	7	22	45	37	50	5	30	58
	PrecStd	288	92	60	60	74	86	94	47	122	60	99	75	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	26.3	27.9	30.2	33.3	36.1	35.7	36.5	34.9	32.5	28.4	26.4
		MCWB	20.4	20.8	20.5	22.2	23.2	25.0	25.8	25.8	25.3	23.8	22.0	21.4
	2%	DB	23.6	24.7	26.9	28.9	32.1	34.0	34.2	34.9	33.5	31.1	27.1	24.7
		MCWB	19.6	20.1	20.4	21.9	23.3	24.9	25.8	25.7	25.0	23.7	21.3	21.0
	5%	DB	22.2	23.0	25.7	27.8	31.1	32.8	33.1	33.8	32.4	29.6	25.7	22.9
		MCWB	18.9	19.4	20.1	21.2	23.4	24.9	25.8	25.8	24.6	22.8	20.3	20.0
	10%	DB	20.4	22.0	24.0	27.1	29.9	32.1	32.4	32.7	31.3	28.1	23.9	21.4
		MCWB	18.0	18.9	19.3	20.8	23.0	24.9	25.7	25.7	24.3	22.0	19.4	19.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	22.2	23.0	24.4	25.7	27.4	27.7	27.9	27.2	26.4	24.0	23.3
		MCDB	23.7	24.6	25.8	28.0	30.2	31.2	31.7	31.9	30.5	29.3	26.4	25.0
	2%	WB	20.8	21.3	22.2	23.6	25.1	26.7	27.1	27.4	26.5	25.5	22.8	21.9
		MCDB	22.6	23.6	24.6	26.9	29.3	30.8	31.3	31.4	30.3	28.8	25.2	23.8
	5%	WB	19.6	20.5	21.3	23.0	24.6	26.2	26.7	26.9	25.9	24.6	21.8	20.9
		MCDB	21.3	22.6	23.9	26.0	28.6	30.4	31.0	31.0	29.9	27.9	24.2	22.8
	10%	WB	18.2	19.4	20.5	22.4	24.1	25.7	26.3	26.5	25.4	23.8	20.6	19.4
		MCDB	20.0	21.3	23.2	25.4	28.0	30.0	30.5	30.7	29.5	26.7	23.5	21.0
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.2	10.7	10.6	9.9	9.0	8.8	9.2	9.6	11.5	11.7	10.5
		MCDBR	10.6	10.3	10.2	10.5	10.3	10.2	9.7	10.4	10.5	11.3	11.8	10.6
		MCWBR	7.4	6.6	5.4	5.0	3.8	3.4	3.5	3.6	3.7	5.1	6.6	7.4
	5% WB	MCDBR	9.4	8.4	8.8	8.7	8.7	8.7	8.9	9.2	8.9	9.2	9.9	9.8
		MCWBR	7.6	6.1	5.8	5.1	3.7	3.6	3.7	3.7	3.7	5.0	7.1	7.8
Clear-Sky Solar Irradiance	taub	0.339	0.356	0.379	0.420	0.448	0.473	0.492	0.485	0.445	0.381	0.359	0.351	
	taud	2.503	2.465	2.389	2.277	2.237	2.212	2.193	2.209	2.290	2.444	2.454	2.474	
	Ebn at Noon	894	909	907	876	849	822	806	808	831	869	862	859	
	Edh at Noon	90	102	117	135	141	144	147	143	127	102	94	89	
All-Sky Solar Radiation	RadAvg	2.93	3.43	4.68	5.70	6.28	6.14	5.84	5.52	5.08	4.51	3.40	2.62	
	RadStd	0.29	0.49	0.42	0.41	0.36	0.49	0.42	0.48	0.45	0.51	0.33	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+80

Nomenclature: See separate page